

Writing Growth and Decay Models From Data

Answer Key

Algebra 1 Exponential Functions

Quick Answers

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|----------------------------------|------------------------------|
| 1. 3, 12150 | 6. $\frac{21}{20}$, 2552.56 |
| 2. $\frac{1}{2}$, $\frac{5}{4}$ | 7. 312, 115.49 |
| 3. 8240, 10751 | 8. $\frac{4}{5}$, 81.92 |
| 4. 1020, 626.41 | 9. $\frac{6}{5}$, 4478.98 |
| 5. $\frac{7}{10}$, 14.12 | 10. 2, 3, 768 |
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Curriculum

Level: Algebra 1 Exponential Functions

HSF-LE.A.2 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 21 tagged checks.

Common wrong answers

3. If they got 10400: added 3 percent of the starting population ten times, which is linear growth
4. If they got 480: subtracted 15 percent of the original price four times instead of using a decay factor
9. If they got 3300: treated the first difference as a constant and added 300 dollars each month
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1. Bacteria culture.

Step 1. A table is exponential when consecutive outputs have a constant *ratio*. Divide: $150 \div 50 = 3$, $450 \div 150 = 3$, $1350 \div 450 = 3$. The factor is the same every hour, so an exponential model fits. $b = 3$

Step 2. The value at $t = 0$ is the initial amount, so $a = 50$ thousand cells, and the model is $y = 50 \cdot 3^t$ with t in hours and y in thousands of cells.

Step 3. Part (c): substitute $t = 5$. $50 \cdot 3^5 = 50 \cdot 243$. 12150

Reading the answer: 12,150 thousand cells, i.e. about 12.15 million cells. Keep the unit attached — the model outputs whatever the table measured.

2. Medicine in the bloodstream.

Step 1. Ratios again: $40 \div 80 = \frac{1}{2}$, and the same halving holds for the next two rows.

$$b = \frac{1}{2}$$

Step 2. Because $0 < b < 1$, this is decay — each hour keeps half of what was there. With $a = 80$, the model is $y = 80 \left(\frac{1}{2}\right)^t$, y in mg and t in hours.

Step 3. Part (c): $80 \left(\frac{1}{2}\right)^6 = \frac{80}{64}$.

$$1.25 \text{ mg}$$

Half-life language: a factor of $\frac{1}{2}$ per hour means the half-life is exactly one hour, which is why 6 hours leaves $80/2^6$.

3. Town population.

Step 1. A 3 percent yearly increase keeps the original 100 percent and adds 3 percent, so the factor is $b = 1 + 0.03 = 1.03$. Checking one year: $8000(1.03) = 8240$ residents.

$$8240$$

Step 2. With $a = 8000$, the model is $P = 8000(1.03)^t$, t in years from now and P in residents.

Step 3. Part (c): $8000(1.03)^{10} = 8000(1.343916\dots) = 10751.33\dots$, and residents are whole people.

$$10751$$

Why not add 3 percent ten times: that would give $8000 + 10(240) = 10400$, because it charges 3 percent of the *original* 8000 every year. The model charges 3 percent of the current population, which is larger every year.

4. Laptop depreciation.

Step 1. Losing 15 percent leaves 85 percent, so $b = 1 - 0.15 = 0.85$. One year of the model: $1200(0.85) = 1020$ dollars.

$$1020$$

Step 2. With $a = 1200$, the model is $V = 1200(0.85)^t$, V in dollars and t in years.

Step 3. Part (c): $1200(0.85)^4 = 1200(0.52200625) = 626.4075$, rounded to the cent.

$$626.41$$

Decay never reaches zero: the value keeps 85 percent of a shrinking number, so the model approaches 0 without arriving — a fair description of resale value, and a reason not to extend it too many years.

5. Bouncing ball.

Step 1. The rebound factor is one bounce height divided by the height before it: $84 \div 120 = 0.7$. Confirm with the next row: $58.8 \div 84 = 0.7$.

$$b = 0.7$$

Step 2. The drop height is the value at $n = 0$, so $a = 120$ and the model is $h = 120(0.7)^n$, h in cm and n the bounce number.

Step 3. Part (c): $120(0.7)^6 = 120(0.117649) = 14.11788$, to two decimal places.

$$14.12 \text{ cm}$$

Common slip: using $41.16 \div 58.8$ is fine — it is also 0.7 — but dividing the *earlier* height by the later one gives $1.428\dots$, which would say the ball bounces higher each time. Always divide later by earlier.

6. Savings account, two readings.

Step 1. Two years of growth multiply by b twice, so the readings give $2000b^2 = 2205$, hence $b^2 = \frac{2205}{2000} = 1.1025$ and $b = \sqrt{1.1025}$. $b = 1.05$

Step 2. Part (b): $b = 1.05$ means $1 + r = 1.05$, so $r = 0.05$ — a 5 percent annual rate. The model is $B = 2000(1.05)^t$, B in dollars and t in years.

Step 3. Part (c): $2000(1.05)^5 = 2000(1.2762815625) = 2552.563125$, rounded to the cent.

2552.56

Only the positive root is used: $b = -1.05$ also satisfies the equation, but a negative factor would flip the balance's sign every year, which no account does.

7. Algae treatment.

Step 1. Removing 22 percent leaves 78 percent, so $b = 1 - 0.22 = 0.78$. After one week: $400(0.78) = 312$ square metres, which matches the 22 percent removal ($400 - 88 = 312$).

312

Step 2. With $a = 400$, the model is $A = 400(0.78)^w$, A in square metres and w in weeks.

Step 3. Part (c): $400(0.78)^5 = 400(0.2887174\dots) = 115.48697\dots$, to two decimal places.

115.49 m²

Interpretation worth asking for: after 5 weeks a little over a quarter of the original cover remains. The treatment is working, but 22 percent per week never clears the pond completely.

8. Drug with two readings.

Step 1. The readings are 2 hours apart, so the hourly factor is applied twice: $250b^2 = 160$, giving $b^2 = \frac{160}{250} = 0.64$ and $b = \sqrt{0.64}$. $b = 0.8$

Step 2. Part (b): the dose at $t = 0$ is known, so $a = 250$ and the model is $y = 250(0.8)^t$, y in mg and t in hours.

Step 3. Part (c): $250(0.8)^5 = 250(0.32768)$.

81.92 mg

Check against the data: the model must reproduce the reading it was built from — $250(0.8)^2 = 160$ mg, exactly the 2-hour blood test. Substituting the given point back is the fastest way to catch a wrong root.

9. Is the campaign exponential?

Step 1. Differences: $1800 - 1500 = 300$, $2160 - 1800 = 360$, $2592 - 2160 = 432$. Not constant, so no linear model fits. Ratios: $1800 \div 1500 = 1.2$, $2160 \div 1800 = 1.2$, $2592 \div 2160 = 1.2$. Constant ratio means exponential. $b = 1.2$

Step 2. With $a = 1500$, the model is $T = 1500(1.2)^t$, T in dollars and t in months. A factor of 1.2 is a 20 percent monthly increase.

Step 3. Part (c): $1500(1.2)^6 = 1500(2.985984) = 4478.976$, rounded to the cent. 4478.98

The trap this problem sets: the first difference is 300, so a student who stops there predicts $1500 + 6(300) = 3300$ — more than a thousand dollars low. Constant differences and constant ratios are different patterns, and only the ratio test survives all three rows here.

10. Challenge: mould colony, neither reading at day 0.

Step 1. Day 5 minus day 2 is 3 days, so the factor is applied three times: $12b^3 = 96$, giving $b^3 = 8$ and $b = \sqrt[3]{8}$. $b = 2$

Step 2. Part (b): a is the area at day 0, and the day 2 reading is $ab^2 = 12$. Working backwards, $a = \frac{12}{2^2} = \frac{12}{4}$. $a = 3$

Step 3. The model is $A = 3 \cdot 2^t$, A in square centimetres and t in days. Check it against both photographs: $3 \cdot 2^2 = 12$ and $3 \cdot 2^5 = 96$.

Step 4. Part (c): $A = 3 \cdot 2^8 = 3(256)$. 768 cm^2

Why this one is harder: when neither reading is at $t = 0$, the ratio still gives b — but a has to be recovered by dividing one reading by b raised to that reading's own time. Students who set $a = 12$ get every prediction four times too large.